

Numerical Simulation of Learning Dynamics in Multimodal RL

A tunable diversity correction for outcome-level mode collapse

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Background

Base Paper: The Collapse Mechanism

Sinha, Elango and Liu (2026), *arXiv:2601.21669*.

Many RL tasks have **several equally good answers**: valid reasoning chains, distinct drug candidates, different correct programs. We want the policy to keep all of them, not pick one.

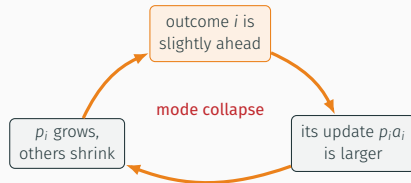
The standard objective is expected return

$$J(\theta) = \mathbb{E}_{o \sim p_\theta} [r(o)] = \sum_i p_i r_i.$$

Under gradient flow the paper proves (Thm. 3.1)

$$\dot{z}_i = p_i a_i, \quad a_i = r_i - \bar{r}.$$

An outcome's update is scaled by **how likely it already is**. That is a rich-get-richer loop, and it runs even when two outcomes have identical reward.



Collapse is a property of **the objective itself**. Entropy bonuses and KL penalties push against it, but they do not remove the cause.

Base Paper: Inverse Probability Scaling

Inverse Probability Scaling. Divide the reward by the probability of the outcome that produced it:

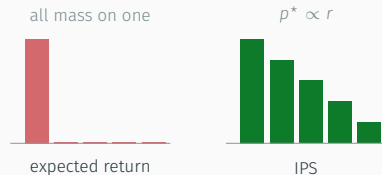
$$\tilde{r}(o) = \frac{r(o)}{p_{\theta}(o)}.$$

The $1/p$ cancels the p_i multiplier, so (Thm. 4.1, Cor. 4.2)

$$\dot{z}_i = r_i - p_i \sum_k r_k \implies p_i^* = \frac{r_i}{\sum_k r_k}.$$

In practice $p_{\theta}(o)$ is unknown. IPS-GRPO estimates it from a group of G sampled rollouts and clips (their Eq. 9):

$$\tilde{r}(o_g) = \frac{r(o_g)}{\max\{\hat{p}(o_g), \varepsilon\}}.$$



Left open by the paper:

1. Only two settings exist: plain (collapse) or exact proportionality.
2. The group size G and clip ε are ablated empirically only (their Table 4).
3. Their own limitation section: finite-group estimates “can be noisy, requiring probability clipping that introduces a bias-variance trade-off”. No analysis is given.

Project Overview

Problem Statement and Extension

The paper gives one fixed correction. We asked three questions it does not answer.

1. Can the correction be tuned?

Replace r/p by r/p^α . Redoing the paper's own gradient calculation gives

$$\dot{z}_i = r_i p_i^{1-\alpha} - p_i \sum_k r_k p_k^{1-\alpha},$$

$$p_i^* \propto r_i^{1/\alpha}$$

One knob spans collapse, the paper's result, and near-uniform.

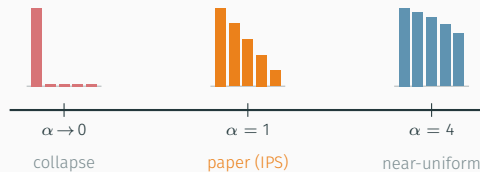
2. What does a finite group cost?

Derive the effect of G and ε instead of ablating it.

3. How wrong is the estimate $\hat{p}$?

The error analysis the paper lists as an open limitation.

stationary policy p^* for $r = (5, 4, 3, 2, 1)$



α becomes a **diversity dial**: the same algorithm, one extra exponent.

System Model

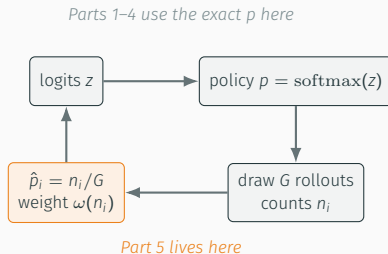
An **outcome-selection bandit**: K outcomes, fixed rewards $r_i > 0$, policy stored as logits and squashed by softmax,

$$p_i = \frac{e^{z_i}}{\sum_k e^{z_k}}.$$

Nothing else is in the model. No network, no exploration schedule, no critic. So any collapse we see has **exactly one possible cause**, which is the point the paper argues.

Two objects are studied throughout:

- the **idealised flow** $\dot{z} = f(z)$, an ODE in K variables;
- the **sampled update**, one noisy step built from G draws.



$$\hat{g}_i = \hat{p}_i r_i \omega(n_i) - p_i \sum_k \hat{p}_k r_k \omega(n_k), \quad z \leftarrow z + h \hat{g}$$

Work Breakdown

Part	Focus	Key content
1	Collapse baseline, $\alpha = 0$	Integrate the flow with Euler and RK4, verify orders, show the exact tie is provably flat, and that sampling noise alone still collapses it
2	The α knob	Stationary law $p^* \propto r^{1/\alpha}$, linearisation and eigenvalue spectrum, convergence rates, explicit-method stability limits, first finite- G ceiling
3	Solving $\dot{z} = 0$ directly	Bisection, secant and Newton on three equivalent formulations; convergence orders, basins, conditioning; nested solver for the finite- G fixed point
4	Multi-parameter atlas	19,125 stationary points over $(\alpha, G, \varepsilon, \rho)$, support and entropy for general K , and whether interpolation can replace solving
5	The estimator itself	Bias and variance of the clipped weight, Richardson extrapolation, golden-section search, power and QR methods, moving averages

One shared library (`src/`), one script per part (`experiments/`), one write-up per part, and 68 unit tests.

Part 1: Collapse Baseline

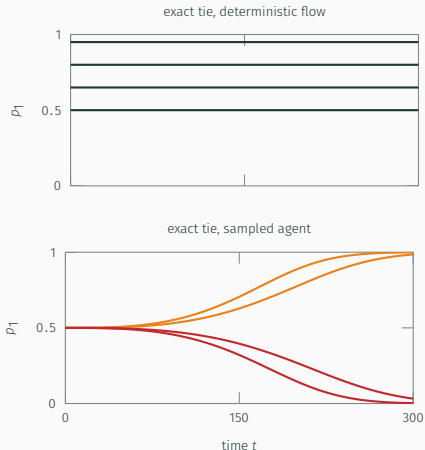
Part 1: Objective

Before correcting collapse we had to reproduce it, and to be precise about a claim the paper states informally.

The paper says collapse happens “even with exactly equal rewards”. For the **deterministic** flow that is not literally true: if all r_i are equal then $a_i \equiv 0$, so $\dot{z} \equiv 0$ and nothing moves at all.

What actually drives collapse under a tie is **sampling noise**. A finite group never sees the exact mean reward, so the realised advantage is never zero, and the p_i multiplier then amplifies it.

So Part 1 runs both and separates them.



Same rewards, same starting policy. Only the noise differs.

Part 1: Numerical Methods

Solution of ODEs. The flow is a coupled system in K unknowns,

$$\dot{z}_i = r_i p_i - p_i \sum_k r_k p_k.$$

Integrated with Euler $z_{n+1} = z_n + hf(z_n)$ and classical RK4

$$z_{n+1} = z_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Truncation and round-off error. Empirical order from a step-size sweep,

$$\text{order} = \text{slope of } \log E \text{ vs } \log h.$$

Below $h \approx 0.025$ the RK4 error stops falling: the truncation term has dropped under the round-off floor.

Least-squares regression. Collapse time against reward gap, fitted in log-log space,

$$\log t_{\text{col}} = m \log \Delta + c \Rightarrow t_{\text{col}} \approx 53.94 \Delta^{-1.00}.$$

The same fit gives the group-size law $t_{\text{col}} \propto G^{1.07}$.

Random number generation. The sampled agent draws outcomes by **inverse transform sampling**: build the CDF of p , draw $u \sim U(0, 1)$, take

$$o = \min\{i : \sum_{k \leq i} p_k \geq u\}.$$

Visualisation. Semi-log and log-log axes, since both the transition and the fitted laws span decades.

Part 1: Results and Analysis

The tie is provably flat. Four different starting policies, $T = 300$:

$$\max_t |\Delta p_1| = 1.11 \times 10^{-16}$$

which is machine epsilon, not a slow drift. This **sharpens** the paper's claim rather than contradicting it.

Noise alone collapses it. From an exactly symmetric start ($z_0 = 0$), $K = 5$, 200 seeds:

outcome	O1	O2	O3	O4	O5
won	18.5%	22.5%	19.0%	19.0%	21.0%

All inside the 95% band [14.5, 25.5]% around $1/K$. The winner is arbitrary.

Two clean scaling laws.

$$t_{\text{col}} \approx 53.94 \Delta^{-1.00}, \quad R^2 = 0.99999994$$

$$t_{\text{col}} \propto G^{1.07}, \quad R^2 = 0.974$$

A bigger group delays collapse roughly linearly. That is a statement about a real GRPO hyperparameter.

Integrators behave.

	order	error at $h = 0.05$
Euler	1.00	1.4×10^{-4}
RK4	3.99	5.1×10^{-8}

Reported honestly: we looked for an Euler instability here and found none. The right-hand side is bounded and self-damping at $\alpha = 0$, so no step size can overshoot. Part 2 is where instability appears.

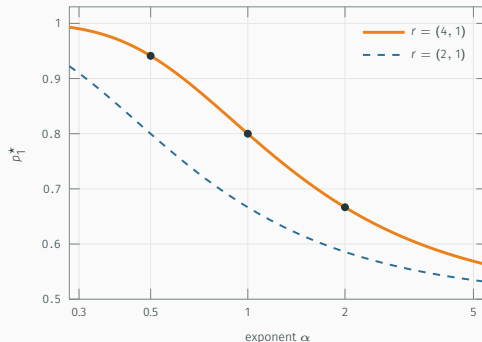
Part 2: The Diversity Exponent

Part 2: Objective

Test every prediction of the α generalisation, first on the exact ODE and then on the sampled agent.

- Does the flow really settle at $p^* \propto r^{1/\alpha}$?
- *How fast* does it get there, and can we predict the rate before running anything?
- Which step sizes are safe for an explicit method?
- Does a finite group change the answer?

The last question produced the result that the rest of the project is built on.



Marked points: 94/6, 80/20, 67/33. RK4 endpoints land on the analytic curve to 7.5×10^{-15} .

Part 2: Numerical Methods (1/2)

Modelling with differential equations. Stationary points come from setting $\dot{z}_i = 0$, so $r_i p_i^{-\alpha}$ must be the same constant for every i :

$$p_i^* \propto r_i^{1/\alpha}.$$

Eigenvalue decomposition. Linearise at p^* . With the softmax derivative $C = \text{diag}(p^*) - p^* p^{*\top}$, perturbing the flow gives

$$J^* = -\alpha \|r\|_{1/\alpha} C.$$

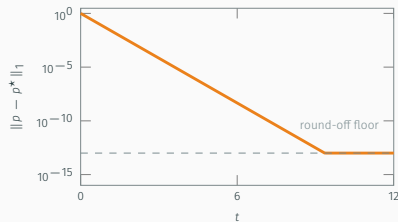
J^* is symmetric negative semidefinite. Its nonzero eigenvalues are the convergence rates

$$\lambda_j = \alpha \|r\|_{1/\alpha} \mu_j,$$

with μ_j the nonzero eigenvalues of C . One zero eigenvalue survives, along $\mathbf{1}$, because shifting all logits leaves p unchanged.

Least-squares regression. Fit $\ln \|p(t) - p^*\|_1$ against t inside a window past the nonlinear transient and above round-off. The slope is the measured rate, checked against $\lambda_{\min}$ above.

Round-off error. The fit window is chosen as $[10^{-12}, 10^{-6}]$ precisely because below 10^{-12} the error is round-off, not dynamics.



Part 2: Numerical Methods (2/2)

Stability of explicit ODE methods. A linear mode with rate λ is stable only if

$$h\lambda_{\max} < 2 \text{ (Euler)}, \quad h\lambda_{\max} < 2.7853 \text{ (RK4)}.$$

The RK4 constant is the real root of $x^3 - 4x^2 + 12x - 24 = 0$, which we obtain by **bisection** rather than quoting it.

Root finding: bisection. The critical step size is found by bisection *on the actual nonlinear iteration*: shrink or grow h until a 10^{-4} perturbation of z^* stops decaying. It matches the linear theory to 2.6×10^{-4} (Euler) and 1.3×10^{-4} (RK4).

Random number generation and simulation. The sampled agent is one noisy Euler step per iteration. Group frequencies obey

$$G\hat{p}_i \sim \text{Binomial}(G, p_i).$$

Taking the expectation of that update replaces $p_i^{-\alpha}$ by an **effective weight**

$$w_G(p) = \frac{\mathbb{E}[\hat{p}^\alpha(n)]}{p},$$

whose stationary condition $r_i w_G(p_i) = S$ is solved by bisection.

Model validation. Every Monte-Carlo run is compared against that mean-field prediction, which is the $h \rightarrow 0$ limit of the same update.

Part 2: Results and Analysis

The knob works, and the rate is predictable. Fitted rate against $\lambda_{\min}$ from the Jacobian: relative error $\leq 1.7 \times 10^{-5}$ over 27 runs ($K = 2$).

Both ends of the dial are numerically hard.

α	$\lambda_{\min}$	$\lambda_{\max}$	ratio
0.1	5.8×10^{-8}	0.089	1.5×10^6
1	1.15	4.53	3.9
5	7.2×10^3	9.3×10^3	1.3

Small α is stiff (huge spread of time scales). Large α makes every rate large, so stability, not accuracy, sets h . Part 1's step $h = 0.05$ becomes unstable above $\alpha = 2.06$ for Euler.

$\alpha = 0$ is a **singular limit**: the approach is algebraic, $p_2 \sim 1/(2\Delta t)$, fitted tail slope -1.007 , with no exponential rate at all.

The result the project is built on. A group of G sees a rare outcome at most once, at frequency $1/G$. So the inverse-probability weight is **capped**:

$$w_G(p) \rightarrow \min\{G, 1/\varepsilon\}^\alpha \quad \text{as } p \rightarrow 0.$$

The minority outcome survives only if

$$\alpha > \alpha_c = \frac{\ln(r_1/r_2)}{\ln \min\{G, 1/\varepsilon\}}$$

G	4	8	16	32	64
$\alpha_c, r = (4, 1)$	1.0	0.667	0.5	0.4	0.333

At $G = 4$ the paper's own $\alpha = 1$ fails: long-run $p_1 = 0.9965$ instead of 0.8. Monte Carlo tracks the mean field to 0.013 worst case.

Part 3: Root Finding Methods

Part 3: Objective

Part 2 reached p^* by integrating for many e-folding times. Part 3 solves $\dot{z} = 0$ **directly** and checks the root against the analytic answer.

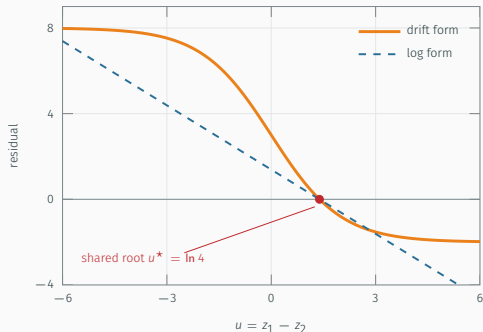
The same condition can be written three ways:

$$\text{drift: } F_i = \dot{z}_i$$

$$\text{balance: } F_i = r_i p_i^{-\alpha} - r_K p_K^{-\alpha}$$

$$\text{log: } F_i = \ln \frac{r_i}{r_K} - \alpha(z_i - z_K)$$

They share the same root. Dividing by p_i removes the very multiplier the paper blames for collapse, and it turns out to matter for Newton too.



The drift saturates at both ends, so a Newton step taken from far out lands even further away. The log form is a straight line, so Newton solves it in one step.

Part 3: Numerical Methods

Root finding. All three from the syllabus, on the same problem:

bisection: halve a sign-change bracket

$$\text{secant: } u_{n+1} = u_n - F_n \frac{u_n - u_{n-1}}{F_n - F_{n-1}}$$

$$\text{Newton: } u_{n+1} = u_n - F(u_n)/F'(u_n)$$

Measured order from the least-squares slope of $\ln e_{n+1}$ against $\ln e_n$.

Systems of equations. For $K > 2$, Newton solves

$$J(y) \delta y = -F(y)$$

by Gaussian elimination each iteration. J is the analytic Jacobian, not a finite difference.

Conditioning, and a gauge. Shifting every logit leaves $\dot{z}$ unchanged, so $J1 = 0$ and the full $K \times K$ system is **singular** (smallest singular value 7.1×10^{-17}). Fixing $z_K = 0$ removes it. At the root

$$J_{\text{balance}} = -\alpha \|r\|_{1/\alpha} I, \quad J_{\log} = -\alpha I,$$

so both have condition number exactly 1.

Nested root finding for finite G . Inner: invert the monotone $w_G(p_i) = S/r_i$ for each outcome. Outer: choose S with $\sum_i p_i(S) = 1$. Both levels are monotone with a guaranteed bracket, so bisection is safe and safeguarded Newton is fast.

Part 3: Results and Analysis

Textbook orders, confirmed. $K = 2$, $\alpha = 1$, root $u^* = \ln 4$:

method	iterations	order
bisection	56	linear (0.502)
secant	9	1.61
Newton (drift)	5	2.00
Newton (log)	1	exact

How you write the equation decides the basin.

Newton on the drift form converges from only 17% of starts at $\alpha = 1$, but from 100% once $\alpha > 1.113$. On the balance form it converges from 100% of starts at every α .

Backtracking does not rescue it: for $\alpha < 1$ every simplex vertex is a zero of the drift “at infinity”, and a line search is pulled into them.

Nested solver: Newton pays off. Over

$G \in \{4, 16, 64\} \times \alpha \in \{0.5, 1, 2\}$:

	outer iters	total work
bisection / bisection	43–55	4,506–13,769
Newton / Newton	6–16	225–1,703

Same answer to 9.8×10^{-14} , with 8 to 20 times less work.

A new result for $K > 2$. The two-outcome threshold understates the requirement. At $G = 16$,

$r = (5, 4, 3, 2, 1)$:

α_c for	02	03	04	05
$K = 5$ mean field	0.0805	0.203	0.419	0.886
pairwise formula	0.0805	0.184	0.330	0.580

Some outcomes are lost at every α : here the weakest needs $G \geq 6$ before any exponent can keep it.

Part 4: Parameter Atlas

Part 4: Objective

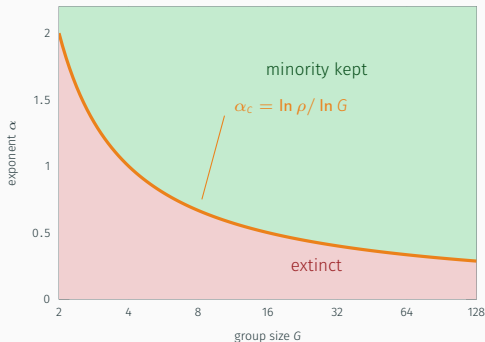
Parts 2 and 3 varied one or two parameters. Part 4 asks whether the survival boundary really holds **jointly** over all four, and whether a grid can stand in for solving.

The predicted boundary collapses to a single signed margin

$$m = \alpha \log \min\{G, \frac{1}{\varepsilon}\} - \log \rho,$$

with $\rho = r_1/r_2$. Survival should be exactly $m > 0$, independent of where in the four-dimensional box you are.

19,125 stationary points were solved to test that one inequality.



One slice ($\rho = 4$, $\varepsilon = 10^{-3}$) of a four-dimensional atlas.

Part 4: Numerical Methods

Interpolation. The atlas is a tensor-product grid in $(\log \alpha, \log \rho, \log \varepsilon)$. Inside one cell, with $t_d \in [0, 1]$ the fractional position along axis d , the **multilinear** interpolant is the weighted sum over the cell's 2^D corners:

$$\hat{f}(\mathbf{x}) = \sum_{\mathbf{c} \in \{0,1\}^D} f(\mathbf{x}_{\mathbf{c}}) \prod_{d=1}^D t_d^{c_d} (1 - t_d)^{1-c_d}.$$

This is the D -dimensional form of the linear interpolation in the syllabus, and it is exact at nodes and for any function affine in each coordinate separately.

Approximation error and order. Refine the grid and fit the slope of $\log(\text{RMSE})$ against $\log(\text{intervals per axis})$.

Root finding, vectorised. All 19,125 $K = 2$ solves run as one batched bisection with 64 deterministic halvings, and 40 of them are re-checked against the original scalar solver.

Model validation and verification. The script refuses to write a summary unless every check passes: phase classification, stationarity residual, normalisation, node exactness, and monotone refinement.

Where the theory says interpolation must degrade. The response is continuous but has derivative kinks at

$$m = 0 \text{ (extinction),} \quad \varepsilon = 1/G \text{ (clip activates),}$$

so cells crossing those surfaces cannot keep second-order behaviour. We measure the two regions separately instead of quoting one rate.

Part 4: Results and Analysis

The boundary holds everywhere. Over $51 \times 15 \times 5 \times 5 = 19,125$ points:

$$\text{misclassifications} = 0/19,125$$

with worst stationarity residual 1.18×10^{-15} .

Two separate ceilings. ε can bind before G does. At $\alpha = 1, \rho = 4$:

p_2 at	$G=6$	$G=16$	$G=64$	$G=256$
$\varepsilon = 10^{-3}$	0.115	0.195	0.200	0.200
$\varepsilon = 0.3$	0	0	0	0

With $\varepsilon = 0.3$ the clip caps the boost at $3.33 < 4$, so **no group size whatsoever** rescues the minority.

Group size must grow with the problem. Smallest G that keeps every outcome at $\alpha = 1$:

K	spread 1.25	spread 32
2	2	33
5	3	54
10	4	96

A best-versus-worst ratio alone does not predict this once several strong outcomes compete for mass.

Interpolation audit. 800 held-out off-grid points:

region	RMSE (17^3)	order
smooth	2.5×10^{-3}	1.58
near a kink	9.7×10^{-3}	1.03

So a grid is fine for exploring, but survival decisions near $m = 0$ need a real solve. The margin m is useful twice: as the criterion, and as a warning flag.

Part 5: Estimator Error Analysis

Part 5: Objective

Parts 1 to 4 assume the update knows p . It does not. It has G samples, and it needs $p^{-\alpha}$.

This is the paper's own stated limitation, and it is a pure **error-analysis** problem: bias against variance, with a knob (ϵ) that trades one for the other exactly like a step size trades truncation against round-off.

We compare the paper's clipped rule against three alternatives, and ask which trade the *learning dynamics* actually notice.

one group of $G = 16$ draws:

A A B A C B A D B A C B A C B A

counts n_i : $n_A=7$ $n_B=5$ $n_C=3$ $n_D=1$ $n_E=0$

weight $\omega(n_i)$: 2.3 3.2 5.3 16 **1000**

one outcome missed,
so $\omega = \epsilon^{-1}$

Without the clip that entry is ∞ . With it, one unlucky group hands a rare outcome a weight 60 times larger than anything else in the same group.

Part 5: Numerical Methods (1/2)

Truncation error via Taylor expansion. Expand $f(x) = x^{-\alpha}$ about p with the random step $\hat{p} - p$, take expectations, and use the binomial central moments ($\mathbb{E}[\hat{p}] = p$, $\text{Var} = pq/G$):

$$\mathbb{E}[f(\hat{p})] = f(p) + \frac{1}{2}f''(p)\frac{pq}{G} + O(G^{-2}),$$

which gives the relative bias and standard deviation

$$b = \frac{\alpha(\alpha+1)(1-p)}{2Gp}, \quad s = \alpha\sqrt{\frac{1-p}{Gp}}.$$

The small parameter is Gp , the expected number of hits, not G .

Golden-section search. $\text{MSE}(\varepsilon)$ falls then rises, with kinks at every $\varepsilon = n/G$, so a derivative-free bracketing minimiser is the right tool. It shrinks the bracket by $1/\phi = 0.618$ per evaluation.

Richardson extrapolation. Equation above is an expansion in $1/G$ with G -independent coefficients, so the standard combination applies. Splitting the group into halves of size $M = G/2$:

$$\omega_R = 2f\left(\frac{n_A+n_B}{G}\right) - \frac{1}{2}\left[f\left(\frac{n_A}{M}\right) + f\left(\frac{n_B}{M}\right)\right]$$

raises the bias order from 1 to 2. Because $\hat{p}_A + \hat{p}_B = 2\hat{p}$, the first-order noise term is unchanged, so the variance is free.

Part 5: Numerical Methods (2/2)

Eigenvalue decomposition: power method and QR. A moving average over training steps turns the first-order relaxation of Part 2 into a **second-order** system. With $\kappa = \beta/h$,

$$x'' + \kappa x' + \kappa \alpha \|r\|_{1/\alpha} Cx = 0 \iff s^2 + \kappa s + \kappa \lambda_j = 0.$$

The λ_j need the eigenvalues of C , which we compute with the **power method** (with Hotelling deflation) and the **QR algorithm**, both written out and checked against a library.

ODE stability, again. The quadratic rings when $\kappa < 4\lambda$ and decays fastest at $\kappa = 4\lambda$.

Round-off error. The closed form derived for $\alpha = 1$,

$$w_G(p) = \frac{1 - (1 - p)^G}{p},$$

cancels catastrophically for small p . Evaluating it as $-\expm1(G \log1p(-p))/p$ keeps the digits:

p	direct	guarded
10^{-11}	8.3×10^{-8}	2.1×10^{-15}
10^{-13}	3.1×10^{-4}	2.3×10^{-15}

Two more round-off lessons we hit for real: Gram-Schmidt QR silently breaks on the singular C , and the power method must stop on the residual $\|Ax - \lambda x\|$, not on λ , because the Rayleigh quotient is accurate to the *square* of the vector error.

Part 5: Results and Analysis

An exact optimum for the clip. Differentiating $\text{MSE}(\varepsilon)$ piecewise, the sign of the derivative is the sign of $p^{-\alpha} - \varepsilon^{-\alpha}$, so

$$\boxed{\varepsilon^* = p}$$

exactly, for every α , G and p . Golden-section search returns $\varepsilon^*/p = 1 \pm 7 \times 10^{-9}$, which is the $\sqrt{\epsilon_{\text{mach}}}$ floor of any derivative-free minimiser.

Since ε is one global constant but ε^* is per-outcome, **no single clip suits a majority and a minority outcome at once**. That is an honest explanation of why the paper's own ε ablation has no clear winner.

Richardson works, but must be guarded. Bias order $1.03 \rightarrow 2.10$, variance unchanged. Naive Richardson returns a **negative** weight for a singleton (-472 at $G = 16$), which is a sign-flipped reward.

The result we did not expect. The dynamics do not see $\mathbb{E}[\omega(n)]$. They see a size-biased version, $w_G(p) = \mathbb{E}[\omega(1+B)]$ with $B \sim \text{Bin}(G-1, p)$. Expanding it,

$$w_G(p)p^\alpha - 1 = \frac{\alpha(\alpha-1)(1-p)}{2Gp} + O(G^{-2}),$$

which **vanishes at $\alpha = 1$** , the paper's own exponent. In fact there it is exact: $w_G(p) = \frac{1-(1-p)^G}{p}$.

So correcting the estimator's bias *destroys* that cancellation. Training ℓ_1 to the ideal p^* , $K = 5$:

G	clip (paper)	guarded Richardson
8	0.2439	0.2172
16	0.1092	0.0703
64	0.0016	0.0294

The better estimator trains worse once G is large.

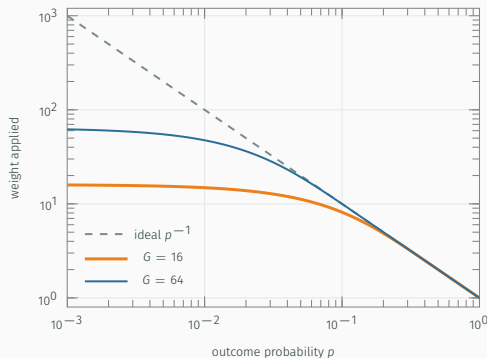
Overall Findings

Comparative Analysis

Claim	Paper	Our work
Expected return collapses	Proved for the idealised flow	Reproduced, and sharpened : under an exact tie the deterministic flow does not move at all (10^{-16}); the collapse is driven by sampling noise
IPS gives $p^* \propto r$	Proved and demonstrated	Reproduced to 7.5×10^{-15} , and generalised to $p^* \propto r^{1/\alpha}$
Group size G and clip ε matter	Ablated empirically, their Table 4	Derived : survival needs $\alpha > \ln \rho / \ln \min\{G, 1/\varepsilon\}$, verified at 0/19,125 misclassifications
Clipping brings a bias against variance trade-off	Stated as a limitation	Quantified : exact bias and variance, the optimal clip $\varepsilon^* = p$, and three alternative rules compared

We did not reproduce the paper's LLM or molecular experiments, which need infrastructure outside this course. What we reproduced is the *mechanism*, in a model where nothing else can explain the result.

Cross-Cutting Result: The Finite-Group Ceiling



A finite group cannot apply an unbounded correction.

The ideal weight $p^{-\alpha}$ grows without limit as an outcome gets rarer. What a group of G can apply saturates, because a rare outcome is seen at most once, at frequency $1/G$:

$$w_G(p) \rightarrow \min\{G, 1/\varepsilon\}^\alpha.$$

Every part meets this ceiling from a different direction:

- Part 2 finds it, and gets α_c for $K = 2$
- Part 3 shows $K > 2$ needs a strictly larger α
- Part 4 verifies it over 19,125 points
- Part 5 shows only $\omega(1)/\omega(G)$ decides it

At $\alpha = 1$ the curve above is exactly $(1 - (1 - p)^G)/p$, proved in Part 5.

Conclusions and Limitations

What a practitioner should take away

- Pick G first. If $\min\{G, 1/\epsilon\} < r_{\max}/r_{\min}$, no exponent and no amount of training will keep the weak outcomes.
- Do not set ϵ above $1/G$ without checking: it introduces a second, independent ceiling.
- For K outcomes, the pairwise threshold is optimistic. At $G = 16$ the weakest of five outcomes needs $\alpha = 0.886$, not 0.580.
- A moving average buys the variance of a $19\times$ larger group at $\beta = 0.1$ with no extra rollouts, but it makes training ring if $\beta < 4h\lambda$.

What we are not claiming

- Everything is the outcome-selection bandit. Our update is group-REINFORCE without GRPO's advantage standardisation or PPO clipping, so the conclusions are about the *weight rule*, not the whole algorithm.
- The α -matched offset we derived is second order, not exact, and it costs dynamic range when $\alpha > 1$.
- A per-outcome moving average assumes a small, enumerable outcome set. That is true here and false for the paper's LLM setting.
- The global convergence proof for general α is not attempted. We state the fixed point analytically and demonstrate convergence numerically from many initial conditions.

Thank you

Questions